■ NEUROSCAN AI

DIAGNOSTIC REPORT

Brain MRI Analysis — AI Assisted

Report ID:	RPT-CC8DCC96	**Generated:**	2026-04-13 14:10:33 IST
Patient:	jo	**Study Date:**	April 13, 2026
Patient ID:	123	**Modality:**	MRI Brain (AI Inferred)
Model:	BrainTumorAI v2.0 Ensemble	**Model AUC:**	0.9996 | Acc 98.78%

PRIMARY FINDING

NO TUMOR

Diagnosis: NO TUMOR
Confidence: ■■■■■■■■■■■■■■■■■■■■ 77.1%
Risk Level: ■ LOW
Processing: 16.95s (TTA 6-view ensemble)
Method: EfficientNet-B3 + ResNet50+CBAM + DenseNet121

DIFFERENTIAL DIAGNOSIS

Class	Probability	Bar Chart	Notes
Glioma	**5.8%**	■■■■■■■■■■■■■■■■	
Meningioma	**9.5%**	■■■■■■■■■■■■■■■■	
No Tumor	**77.1%**	■■■■■■■■■■■■■■■■	*← Primary*
Pituitary	**7.6%**	■■■■■■■■■■■■■■■■	

ENSEMBLE AGREEMENT

Model	Confidence	Bar
EfficientNet-B3	**91.7%**	■■■■■■■■■■■■■■
ResNet50+CBAM	**42.9%**	■■■■■■■■■■■■■■
DenseNet121	**91.7%**	■■■■■■■■■■■■■■
Ensemble (TTA)	**77.1%**	**← FINAL**

IMAGING EVIDENCE

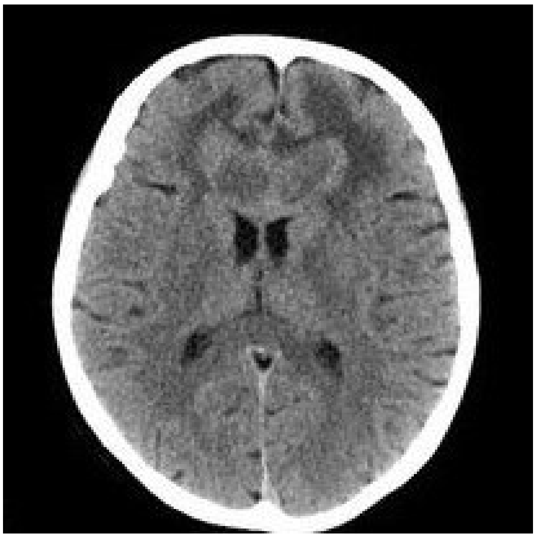

Figure 1: Original MRI scan

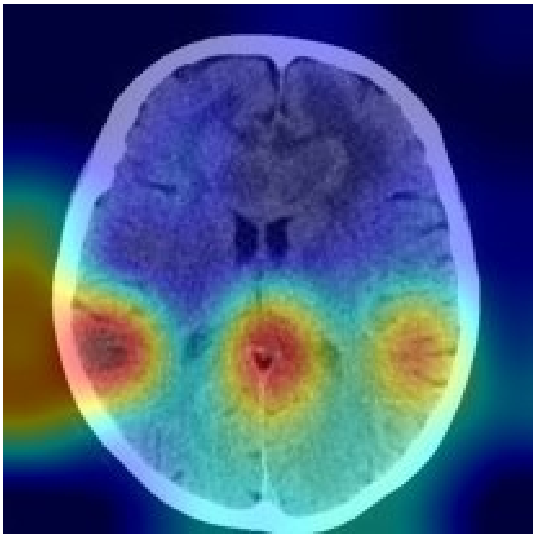

Figure 2: Grad-CAM activation overlay

GRAD-CAM ANALYSIS PANEL

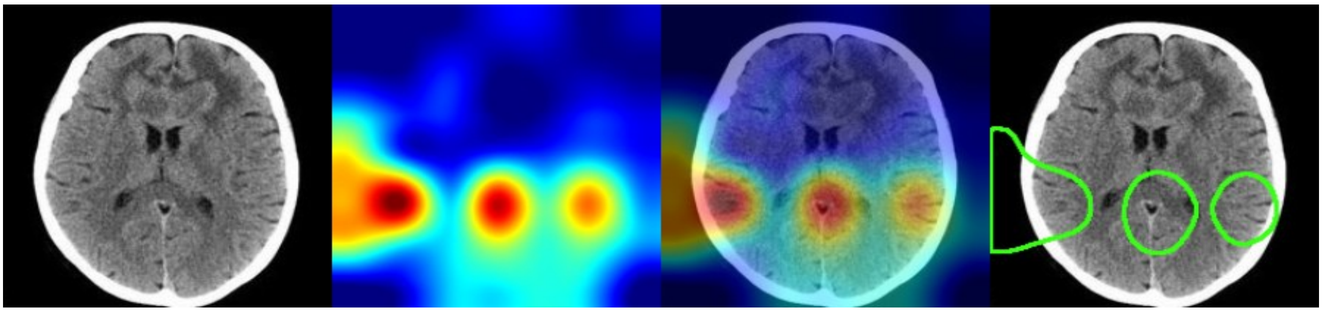

Figure 3: (Left→Right) Original MRI | Activation Map (Jet) | GradCAM Overlay | Tumour Boundary (threshold > 0.5)

■ Red areas = high tumour probability ■ Yellow = moderate ■ Blue = low activation

TUMOUR LOCALISATION (GRAD-CAM ANALYSIS)

Parameter	Value	Notes
Estimated Area	16.7% of visible scan	Moderate
Hemisphere / Location	CENTRAL	Based on centroid x-position
Centroid Position	(75, 140) pixels	Hottest activation pixel centroid
Bounding Region	[0,87] → [178,175]	Pixel coordinates of activation box
Activation Score	67 / 100	Mean activation in region (0=low, 100=high)
GradCAM Computed	4.72s	CPU inference time

CLINICAL INTERPRETATION

No significant intracranial mass lesion identified by AI analysis. The Grad-CAM map shows no focal area of high activation consistent with tumour pathology. Routine clinical follow-up is advised as clinically indicated. If symptoms persist or new neurological signs develop, repeat imaging should be considered. This result does not exclude all pathologies — clinical correlation is essential.

RECOMMENDATIONS

1. Correlate with clinical history and neurological examination
2. No urgent imaging follow-up required based on AI screening
3. Repeat MRI in 6–12 months if symptoms persist
4. Consider EEG if seizures are present

TECHNICAL DETAILS

Parameter	Value
Architecture	EfficientNet-B3 + ResNet50+CBAM + DenseNet121
Inference	TTA (6-view: orig, H-flip, V-flip, rot90/180/270)
Ensemble method	Confidence-weighted average + temperature scaling
GradCAM method	Standard Grad-CAM on last conv layer per backbone
Model AUC	0.9996 (macro one-vs-rest, test set n=1311)
Model Accuracy	98.78%
Sensitivity	98.68% | Specificity: 99.59%
Device	CPU (no GPU detected)
Processing time	16.95s (inference) + 4.72s (GradCAM)

■ ***DISCLAIMER:*** *This report is generated by NeuroScan AI, an AI-assisted screening tool. It is intended for informational and research purposes only and does* ***NOT*** *constitute a medical diagnosis. All findings must be reviewed and confirmed by a qualified radiologist or neurologist before any clinical decision is made. Individual model performance may vary across patient populations and imaging protocols. Reported metrics: AUC 0.9996, Accuracy 98.78%, Sensitivity 98.68%, Specificity 99.59% (test set n=1,311 images). Report generated: 2026-04-13 14:10:33 IST | Report ID: RPT-CC8DCC96*